DRAKEN 2045

PROJECT SALVATOR

Virtual Varanid Substrate · Multi-Scale Coherence Engine · Hardware Stack Specification

A Research Programme Proposal

for the construction of a sheaf-theoretic computational ethology of Varanus salvator komaini,

the minimum-viable hardware substrate for the Draken coherence engine, and
the apex-predator alignment logic for global-scale environment models.

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Abstract

Project Salvator is a research programme to construct a virtual *Varanus salvator komaini* — a computational ethology of the Asian water monitor implemented as a multi-scale dynamical system whose top-level integration layer is the Draken sheaf-theoretic coherence engine. The programme has three deliverables. **(i)** A working *salvator substrate* — a layered software stack from biophysical neuron models through embodied active-inference loops to an 18-layer cellular-sheaf integration manifold computing the coherence metric Γ continuously over time. **(ii)** A *minimum-viable hardware specification* — a development tier (commodity GPU + JAX/PyTorch), a research tier (heterogeneous CPU/GPU/neuromorphic mesh: SpiNNaker 2 / Intel Loihi 2 / BrainScaleS-2), and a deployment tier targeting energy budgets commensurate with biological cognition (≈10–20 W). **(iii)** An *apex-predator logic layer* — a top-level alignment-and-stabilisation function for global-scale world models, whose role is to act as a substrate-invariant boundary condition preventing sub-agent optimisation processes from collapsing the viability of established ecosystems. The biological referent — the varanid hunting protocol, refined for ≳130 Myr and demonstrably substrate-invariant across species, habitat, and body size — is treated as the canonical example of an algorithmic structure that solves the optimisation axiom ◆ $min\ S_sys(t)\ s.t.\ dH/dt \geq 0$ ◆ in the limit of ecological complexity.

This proposal is structured to be readable both by human reviewers and by an autonomous agent swarm tasked with prototyping, programming, and validating the salvator substrate. Every reference is linked. Every algorithmic component is named, formalised, and located in the existing literature. The methodological premises were independently endorsed by the recent commentary of Forli & Yartsev in *Cell* [1] and the follow-up by Yartsev in *Nature Neuroscience* [2], which together provide mainstream scientific cover for the choice of (a) a non-mammalian model species under Krogh's principle, (b) continuous neural-dynamics analysis over trial-averaged paradigms, and (c) the use of single-shot computation as the unit of analysis for ritualised agonistic protocol.

How to read this document

§1 frames the strategic context. §2 develops the theoretical foundation: the 18-layer manifold, the cellular-sheaf formalism, the optimisation axiom, and the active-inference closure. §3 specifies the empirical research programme as eight numbered work packages (WP-1 through WP-8). §4 details the layered software architecture of the salvator substrate. §5 specifies the hardware tiers. §6 develops the apex-predator alignment logic and its mapping onto AI safety. §7 gives the milestone roadmap, outreach matrix, and resource requirements. §8 is the substrate-invariance argument. Annex A is the full reference bibliography with persistent links. Annex B is the agent-swarm tool inventory. Annex C is the verification protocol.

1. Strategic Frame

1.1 The window has opened

In October 2024, Forli and Yartsev published in *Cell* [1] a commentary arguing that the dominant reductionist paradigm in systems neuroscience — head-fixed mice, trial-averaged firing rates, narrowly defined functional categories inherited from classical psychology — has produced a literature that describes the neural correlates of behaviour that no animal in nature actually performs. Quoting their own example, the trained mouse in a forced-choice paradigm is the neural analogue of *a professional golfer*, not of an animal solving the problem of survival. They argue for a paradigm shift toward (a) ethologically grounded behaviour, (b) continuous neural-dynamics analysis, (c) Krogh's principle in species selection, and (d) single-shot computation as a legitimate unit of analysis. In February 2026, Yartsev followed up in *Nature Neuroscience* [2] with the sharper formulation that *natural behaviour is what you do, not just where you are* — environmental realism alone does not guarantee ecological validity, and the anchor must be the species-typical algorithmic structure the brain evolved to execute.

These two papers, taken together, are not citations of Draken — they will not become so — but they are an independent derivation of every methodological premise that Draken §9.8 (the Sheaf Ethology pilot) already assumes. The window for proposing a sheaf-theoretic computational ethology of the varanid clinch is therefore open in a way it was not eighteen months ago. Reviewer-defensibility has shifted: the burden of proof now lies with anyone who insists on trial-averaged head-fixed mouse paradigms for studying ritualised combat protocol, not on the proposer of the alternative.

1.2 Why the varanid

The choice of *Varanus salvator komaini* as the model organism is not contingent. It is the load-bearing structural decision of the entire programme, and it follows from four converging lines of argument.

(i) Krogh's principle, ratified. Forli & Yartsev's commentary [1] and Spataro's 2026 review in *Biological Theory* [3] both argue that monitor lizards constitute an underexplored but exceptionally promising clade for comparative cognition. Spataro proposes that two ecological pressures — expanded home-range size and habitat generalism — plausibly drove the divergence of varanid cognitive sophistication from the rest of Squamata. The empirical literature, while sparse by mammalian standards, is consistent: long-term procedural memory across multi-month hiatuses [4], visual discrimination and reversal learning [5], rapid problem-solving in novel apparatus [6], and the ritualised combat protocol itself, which Horn, Gaulke & Böhme (1994) and Auffenberg (1981, 1988, 1994) have documented as substrate-invariant across the genus.

(ii) Substrate-invariance of the protocol. The varanid bipedal clinch is not a behaviour performed by an individual; it is an emergent algorithmic structure executing through the body of an individual. The protocol is the agent; the individual is the substrate. The same five-node phase graph (Approach → Display → Bluff → Engagement → Resolution) appears in *V. komodoensis* (90 kg, terrestrial), *V. salvator* (20 kg, semi-aquatic), and *V. tristis* (1 kg, arboreal), refined for an estimated 130 Myr of evolutionary continuity. This is the empirical signature of an optimisation process whose endpoint satisfies the Draken axiom

$$\blacklozenge \quad \min S_sys(t) \quad \text{subject to} \quad dH/dt \geq 0 \quad \blacklozenge$$

where *S_sys* is system entropy production and *H* is the Shannon information content of the dyad. The protocol is, in formal terms, a long-time-horizon attractor of the joint dyadic dynamical system, and the restriction maps that compose its phase transitions are conserved across the clade.

(iii) Reptilian neural architecture as a fresh substrate. Tosches et al. (2018) [7] established via single-cell transcriptomics that reptilian and mammalian pallial cell types are largely homologous at the molecular level despite radically different tissue architecture: the mammalian six-layered neocortex is, in their framing, *a mosaic of ancient and new neuronal types* generated by diversification of ancestral gene-regulatory programmes, while reptilian GABAergic interneuron classes already existed in the common amniote ancestor. The dorsal ventricular ridge (DVR) of reptiles performs the same canonical pallial computations as mammalian cortex but does so under a nuclear rather than laminar topology — a different physical implementation of the same computational primitives. This makes the varanid an ideal substrate for testing whether Draken's sheaf formalism captures invariants that hold across implementational divergence rather than artefacts of mammalian cortical layering.

(iv) Metabolic honesty. Reptilian cognition is metabolically gated. A varanid at 20 °C and a varanid at 32 °C are not the same cognitive system; their reaction rates, decision latencies, and even attentional precision are modulated by the Arrhenius gate

$$k(T) = k_0 \cdot \exp[\, -E_a / (R\,T)\,]$$

Mammalian models silently assume $T = 310.15\ K$ and let metabolic budget vanish into a free parameter. For the varanid, T must be treated as an explicit state variable. This is not a complication; it is a feature. Any honest model of cognition must expose its energy budget. Varanids force the issue at the level of the substrate.

1.3 What is being built and why now

The programme aims to deliver, by 2030, a working virtual *salvator* substrate that simulates dyadic agonistic encounters with sufficient fidelity to (a) reproduce empirically observed clinch

outcome distributions from Frýdlová's (Charles University Prague) and Auffenberg's data, (b) compute Γ trajectories that distinguish coherent from collapsing dyadic interactions, and (c) serve as the embodied test-bed for the Draken 18-layer integration manifold operating in closed loop. The scientific deliverable is the substrate; the broader function is to instantiate, in a computable form, the apex-predator alignment logic developed in §6 — a top-level integration function whose role in any large-scale agentic system would be analogous to the role of the apex predator in a trophic cascade [8, 9]: a substrate-invariant constraint that prevents sub-agent optimisation processes from collapsing the viability of the surrounding system.

The current viability of the proposal rests on three converging technological maturities: (1) naturalistic neuroscience methods (DeepLabCut [10], SLEAP [11], CEBRA [12], MoSeq [13]) that make quantitative ethology tractable on commodity hardware; (2) neuromorphic hardware (Intel Loihi 2 / Hala Point [14], SpiNNaker 2 [15], BrainScaleS-2 [16]) that provides physical substrates whose energy-per-spike and asynchronous-event-driven semantics match the biological referent within ≈4 orders of magnitude; (3) the sheaf-theoretic formalism of Hansen & Ghrist [17, 18], which provides a mathematically rigorous account of how local sectional data composes into global coherence via restriction maps and the cellular-sheaf Laplacian. Each was effectively unavailable a decade ago. All three are now production-grade.

2. Theoretical Foundation

2.1 The 18-layer ontological manifold

Draken's base space is a stratified manifold *M* of 18 layers, indexed L01 through L18, spanning physical scales from the quantum field through planetary cognition. The canonical naming, as published at draken.info as of April 2026, is:

Layer	Name	Layer	Name
L01	Quantum Field	**L10**	Social Coordination
L02	Chemical Thermodynamics	**L11**	Economic Cognition
L03	Molecular Assembly	**L12**	National Narrative
L04	Bioelectric	**L13**	Political Structure
L05	Neural Integration	**L14**	Economic Topology
L06	Embodied Cognition	**L15**	Cultural Field
L07	Narrative Self	**L16**	Institutional Morphology
L08	Dyadic Signal	**L17**	Civilisational Memory
L09	Group Cognition	**L18**	Planetary Cognition

The varanid programme operates primarily on layers L01–L09 (with the protocol-as-agent argument extending into L10) and treats L11–L18 as the abstraction levels at which the apex-predator alignment function (§6) becomes interpretable as a global stabiliser.

2.2 Cellular sheaves and the coherence metric Γ

Following Hansen & Ghrist [17], a cellular sheaf F on a graph $G = (V, E)$ assigns to each vertex $v \in V$ a vector space $F(v)$ (the *stalk*, here interpreted as the local section-data of a layer or region), and to each incident edge $v \triangleleft e$ a linear restriction map $F(v \triangleleft e) : F(v) \to F(e)$. A *global section* is a 0-cochain $x \in C^0(G; F)$ satisfying the consistency condition that for every edge $e = uv$, $F(u \triangleleft e)\, x_u = F(v \triangleleft e)\, x_v$. The *coboundary* $\delta : C^0(G; F) \to C^1(G; F)$ measures local discord; the *sheaf Laplacian* is the symmetric positive-semidefinite operator

$$L_F = \delta^* \delta : C^0(G; F) \to C^0(G; F)$$

By the Hodge theorem for cellular sheaves [17, Thm 2.2], $H^0(G; F) = \ker L_F$, i.e. the global sections are exactly the kernel of the sheaf Laplacian. This gives Draken its canonical coherence metric. Define the *normalised coherence*

$$\Gamma(t) = \langle \rho_{i\to j}\, \sigma_i(t),\ \sigma_j(t) \rangle \ / \ (\|\sigma_i(t)\| \cdot \|\sigma_j(t)\|)$$

aggregated over the edges of the relevant subcomplex, where $\sigma_i(t) \in F(v_i)$ is the local stalk-state at time t and $\rho_{i\to j} = F(v_i \triangleleft e_{ij})$ is the restriction map. $\Gamma \in [0, 1]$; $\Gamma \to 1$ corresponds to the system being in or near a global section (high coherence); $\Gamma \to 0$ corresponds to maximal local-global discord. The complementary *narrative self-reference ratio* Ψ — the canonical Draken psychosis metric — is constructed from the relative weight of self-referential vs. environment-referential restriction maps; $\Psi \to 1$ corresponds to the kayfabe trap (*local coherence at the cost of global validity*), $\Psi \to 0$ to environment-grounded representation.

2.3 The optimisation axiom

The Draken optimisation axiom is the dynamical-system constraint

$$\blacklozenge \quad \text{minimise} \ \ S_{sys}(t) \quad \text{subject to} \ \ dH(t)/dt \geq 0 \quad \blacklozenge$$

where $S_{sys}(t)$ is the rate of entropy production of the agent-environment joint system and $H(t)$ is the mutual information the agent maintains about the environment. Verbally: minimise dissipation while preserving or growing world-knowledge. The axiom is structurally equivalent to the free-energy principle of Friston, Parr & Pezzulo [19] — the active-inference functional

$$F[q] = E_q\, [\, \ln q(s) - \ln p(o, s)\,] \ = \ D_{KL}[\, q(s) \,||\, p(s|o)\,] \ - \ \ln p(o)$$

decomposes into accuracy and complexity terms whose joint minimisation is the formal expression of $\min S_{sys}$ subject to $dH/dt \geq 0$. The two formulations are dual: Draken provides the *integration* constraint (what counts as a coherent agent across scales, via sheaf cohomology) and active inference provides the *dynamics* (how the agent updates beliefs and acts to minimise expected free energy). Project Salvator unifies them in a single computational stack.

2.4 The coherence debt and corrective action

The *coherence debt* $K(t)$ is the accumulated departure from viable coherence over time, defined (DRK-121, March 2026) as

$$K(t) = \int_0^t [\, \Psi(\tau) - \Psi_{viable}\,] \cdot w(\tau)\, d\tau$$

where $w(\tau)$ is a contextual weighting kernel. Four canonical correction types are identified: (i) restriction-map repair (lowering Ψ by reconnecting decoupled stalks to environment-referential channels); (ii) stalk-data update (active inference proper, lowering F[q]); (iii) topological surgery (changing the underlying graph G — adding or pruning edges); and (iv) substrate substitution (re-grounding a stalk in a different physical implementation, e.g. moving a function from L05 to

L08). The salvator substrate must be capable of executing all four classes; this is the engineering specification of §4.

2.5 The Grobulator operator G

The *Grobulator* operator *G*, from Swedish *grubbla* (to brood, to ruminate iteratively without a guarantee of convergence), is the meta-evaluative operator of the framework. Formally, *G* takes a candidate state of the manifold *(σ, ρ)* and a candidate update direction, applies the sheaf-Laplacian descent step, evaluates the resulting Γ trajectory across multiple contextual regimes, and returns either an improved state, a refusal to update, or a flag that the configuration is in a *local coherence trap* (Ψ high, Γ locally high but globally inconsistent — the kayfabe signature). The Grobulator is the operator that distinguishes *coherence* from *merely-locally-fitted* representations. It is intentionally non-convergent-guaranteed: convergence on every input would imply a closed world model, which the Grobulator is constructed to refuse.

3. Research Programme: Eight Work Packages

The programme is structured as eight numbered work packages (WPs). Each WP has explicit deliverables, dependencies, and methods. The agent swarm should treat the WP graph as a directed acyclic ordering: WPs 1–3 must complete before WP 4 begins; WP 5 (hardware) runs in parallel from year 1; WPs 6–8 are integration, validation, and the apex-predator alignment proof-of-concept.

WP-1: Behavioural corpus construction

Objective. Build the canonical empirical corpus against which all subsequent simulation will be validated. The unit of analysis is the *dyadic agonistic encounter* — one ritualised combat episode in monitor lizards, decomposed into the five-node phase graph (Approach → Display → Bluff → Engagement → Resolution).

Deliverables.

1. A curated video archive of varanid combat encounters, drawn from (a) the Frýdlová & Frynta corpus at Charles University Prague [20], with formal access request as the priority outreach action; (b) Auffenberg's archival film material; (c) the Mendyk and Horn published-still corpus; (d) wild-camera footage from Bangkok urban populations of *V. salvator macromaculatus* [21] and *V. komodoensis* footage from Indonesian field sites.
2. A SLEAP-trained pose-estimation model [11] for varanid skeletons (15-keypoint rig: snout, nape, mid-back, base of tail, shoulder L/R, elbow L/R, wrist L/R, hip L/R, knee L/R, ankle L/R), with a multi-animal extension following Lauer et al. [10]. DeepLabCut is the secondary alternative; both will be validated. Target accuracy: mAP ≥ 0.90 on held-out frames.
3. An annotated phase-graph segmentation: each frame in each encounter labelled with its node assignment {S_1=Approach, S_2=Display, S_3=Bluff, S_4=Engagement, S_5=Resolution}. Ground-truth annotation by Kai with cross-validation against published ethograms (Horn, Gaulke & Böhme 1994; Auffenberg 1994).
4. A behavioural-syllable decomposition using the MoSeq autoregressive HMM framework of Wiltschko / Markowitz / Datta [13], with syllable durations and transition matrices estimated per individual and pooled across the clade.

Method note. Forli & Yartsev [1] explicitly recommend tracking, decomposition, and classification of behaviour from naturalistic video as the prerequisite for meaningful neural recording. WP-1 is the empirical foundation; without it, every downstream simulation result is unverifiable. Spataro [3] notes that *varanid cognitive research* is held back primarily by the absence of standardised, comparable behavioural data across species — WP-1 is also the contribution that fills that gap.

WP-2: Sheaf-ethology pilot — formal validation of the protocol-as-agent claim

Objective. Demonstrate computationally that the five-node phase graph admits a cellular-sheaf structure with restriction maps that are *substrate-invariant* — i.e. that the same sheaf F (up to isomorphism of stalks) describes encounters in multiple varanid species, despite differences in body size, habitat, and venom composition.

Method.

5. Define the base graph G as the five-node line graph S_1 — S_2 — S_3 — S_4 — S_5. Define stalks as configuration vectors per phase. The Bluff-stage stalk lives in $F(S_3) = \mathbb{R}^4$ with coordinates (display amplitude α, head-elevation θ, bipedal-stance probability β, distance d). The Engagement-stage stalk lives in $F(S_4) = \mathbb{R}^3$ with coordinates (clinch force F_max, energy ratio E_ratio, mass differential Δm).
6. Specify the restriction map $\rho_{D\to Cl} : F(S_3) \to F(S_4)$ as a learned linear-affine map $(\alpha, \theta, \beta, d) \mapsto (F_max, E_ratio, \Delta m)$ projecting out the bluff dimension at α = 0 (no engagement without prior display). Three competing game-theoretic models will be fitted: SAG (Sheaf-Aligned Game), classical hawk-dove, and a sequential-assessment model. The SAG model has shown Γ = 0.928 on preliminary synthetic data.
7. Compute the sheaf Laplacian L_F and the per-encounter Γ trajectory. The substrate-invariance claim is operationalised as: the eigenstructure of L_F computed on encounters from *V. salvator*, *V. komodoensis*, and *V. tristis* should cluster within the same isomorphism class, with restriction-map coefficients that differ only in scale.
8. The *Python pipeline* (already specified in §9.8 of the v4.4 thesis) implements: pose extraction → phase segmentation → stalk-vector construction → sheaf-Laplacian computation → eigendecomposition → Γ trajectory → cross-species comparison. Stack: SLEAP for pose, JAX for differentiable sheaf operations, PySheaf or a custom JAX cellular-sheaf module for the cohomology, NumPy/SciPy for the spectral analysis.

Deliverable. A peer-reviewable result of the form: *the protocol is sheaf-invariant across the clade*, with Γ trajectories distinguishing genuine ritualised combat from broken-protocol encounters (e.g. cannibalistic predation, where ρ_{D→Cl} collapses). Frýdlová's per-dyad data is the load-bearing input; the priority outreach action is therefore the formal data-access request. Target venue: *Animal Cognition* or *Royal Society Open Science*, with a parallel preprint on bioRxiv and arXiv (math.AT).

WP-3: Comparative reptilian neuroanatomy and transcriptomics

Objective. Build the cell-type-resolved circuit map of the varanid pallium, basal ganglia analog, and hypothalamus that will serve as the wiring diagram for WP-4.

Approach.

- Anchor on Tosches et al. (2018) [7] — the canonical single-cell transcriptomic atlas of turtle (*Trachemys*) and lizard (*Pogona*) pallium. The MPIBR/ReptilePallium GitHub repository [7-data] provides raw drop-seq data and analysis scripts. While *Pogona* is not a varanid, the molecular logic — three-layered medial cortex (hippocampal homolog), dorsal ventricular ridge (DVR) excitatory cell types, conserved interneuron classes — is taxonomically conserved across Squamata.
- Cross-reference with Hain et al. (2025) cross-species amniote pallium atlas [22] and Naumann et al. (2015) on the reptilian pallium and the homology debate.
- Compile the AVT/mesotocin neuroanatomy from Wilczynski et al. (2017) *Frontiers in Endocrinology* [23] — the canonical review. AVT (the reptilian/non-mammalian homolog of vasopressin) and mesotocin (the homolog of oxytocin) are confirmed in lizards, with documented effects on aggression, dominance, and (in some species) parental and pair-bond behaviours. These nonapeptides are the hormonal substrate of the dyadic-coupling restriction maps.
- Compile the ectothermy literature: Andrews & Pough (foundational physiology), Seebacher & Franklin on temperature-dependent decision latencies, and the Fry et al. (2006) / Dobson et al. (2019) / op den Brouw et al. (2024) venom corpus already in the Varanvetenskap folder.

Deliverable. A canonical *salvator wiring diagram* specified at the level of (a) named brain regions, (b) approximate neuron counts per region (extrapolated from Pogona where direct varanid data is missing), (c) connectivity matrices (cortico-striatal, cortico-DVR, hypothalamo-pallial), (d) primary neurotransmitter systems with explicit Arrhenius temperature gating, and (e) AVT/mesotocin neuromodulatory fields. Format: a structured JSON specification (machine-readable for the agent swarm) and a parallel narrative document for human reviewers.

WP-4: The salvator substrate — multi-scale software stack

Objective. Implement the working virtual varanid as a layered software stack instantiating layers L01–L09 of the Draken manifold, with the L07 → L09 sheaf integration as the canonical top of the embodied stack.

The detailed architectural specification is given in §4. WP-4's deliverable is the working stack itself: a Git monorepo, Dockerised, with continuous-integration tests, capable of running closed-loop simulations of *V. salvator* agonistic encounters in a synthetic environment, computing Γ continuously, and reproducing (within statistical bounds) the empirical clinch-outcome distributions of WP-1 and WP-2.

WP-5: Hardware substrate — minimum-viable specification

Objective. Specify, prototype, and benchmark the hardware tiers on which the salvator substrate runs. Three tiers are defined; full specification is given in §5.

- **Tier 1 (development):** commodity GPU workstation. NVIDIA RTX 4090 / A6000 class. JAX, PyTorch, NEURON. Sufficient for WP-1 through WP-3 and for prototyping WP-4.
- **Tier 2 (research):** heterogeneous mesh combining (a) GPU cluster for the perceptual/motor neural networks, (b) Intel Loihi 2 / Hala Point board access [14] via Intel Neuromorphic Research Community (INRC) membership for spiking-network L05/L06 layers, (c) optionally SpiNNaker 2 access through the Dresden facility for large-scale spiking simulation. Lava framework for Loihi; PyNN for SpiNNaker.
- **Tier 3 (deployment):** the energy-budget-honest deployment target. Goal: full salvator simulation running at < 100 W sustained, on a neuromorphic-dominant substrate. This is a 5–10-year horizon and is contingent on continued maturation of the Hala Point class of systems.

WP-6: Closed-loop validation and CEBRA-based latent extraction

Objective. Validate that the salvator substrate generates neural-dynamics trajectories whose latent embeddings, extracted via CEBRA [12] or a successor contrastive-learning method, are consistent with biological recordings — to the extent biological recordings are available. Where varanid-specific neural data are absent (which is the current state), the validation falls back to (a) species-typical behaviour reproduction (clinch outcome distributions, syllable transition matrices), and (b) cross-species neural analogues from the Yartsev bat corpus [24, 25] and the Snyder et al. [26] human-experimenter representation studies.

Method. CEBRA in self-supervised mode produces consistent embeddings of high-dimensional time-series across subjects and modalities. Run CEBRA on (i) the simulated salvator neural trajectories, (ii) Steinmetz et al. (2019) [27] mouse brain-wide Neuropixels data, (iii) Forli & Yartsev (2023) [24] bat collective-spatial-behaviour data. Use consistency metrics (Schneider et al. [12]) to assess whether the salvator latent geometry recovers the canonical low-dimensional structure of decision-making and dyadic coupling that is invariant across species.

WP-7: The apex-predator alignment layer

Objective. Demonstrate, in a constrained synthetic environment, that the trained salvator substrate can serve as a *top-level alignment-and-stabilisation function* for a multi-agent simulated ecosystem. The detailed argument is developed in §6. The deliverable is a working proof-of-concept: a sub-agent population (foraging mesopredators, generalist herbivores, opportunistic scavengers) exhibits trophic-cascade-style ecosystem collapse in the absence of the apex-predator constraint, and stabilises in its presence. The salvator substrate plays the apex-predator role; its presence enforces a substrate-invariant boundary condition on the sub-agent optimisation processes.

WP-8: Outreach, integration, and corpus dissemination

Objective. Embed the programme in the existing scientific community via targeted outreach, publish corpus and code under permissive licences, and maintain the agent-swarm-readable instruction set (this document, plus the Annex B tool inventory) as a living artefact.

Outreach matrix (per existing four-tier classification):

- **Tier 1:** Yartsev lab (Berkeley) — strongest methodological alignment after the 2024 + 2026 papers; their bat collective-navigation infrastructure is the closest published precedent for what the Sheaf Ethology pilot wants to do at varanid scale. Chalmers contacts and GU zoology remain the local network.
- **Tier 2:** RISE AI, AI Sweden, and the Templeton World Charity Foundation's Major Transitions in Cognitive Evolution project (which funds Spataro [3]).
- **Tier 3:** Master-level student for the WP-2 pilot computation; Chalmers or GU.
- **Tier 4:** Jakob Hansen (UPenn / co-author on cellular sheaves), Petra Frýdlová (Charles University Prague — varanid combat data), and Chris Clemente (Sunshine Coast / varanid biomechanics, Australia). The Andi Mazingo (ICI) connection from prior outreach remains active.

4. The Salvator Substrate: Multi-Scale Software Architecture

The salvator substrate is a five-tier software stack mapping onto the lower nine layers (L01–L09) of the Draken manifold. Each tier has a defined responsibility, a recommended implementation, and a defined interface with the tier above and below. The architecture is deliberately *hybrid* — no single formalism (spiking nets, deep nets, sheaf cohomology, active inference) covers all scales adequately. The integration is achieved through well-defined data contracts at the tier boundaries.

Tier	Layers	Responsibility	Implementation
T1 Biophysical floor	L01–L04	Ion-channel kinetics, synaptic dynamics, metabolic gating; temperature T(t) as explicit state.	NEURON or NEST with reptilian channel parameters; Arrhenius gate on every rate constant.
T2 Circuit layer	L05–L06	DVR, medial cortex (hippocampal homolog), striatum analog, amygdalar nuclei; AVT/MT neuromodulation.	JAX/PyTorch sparse RNN with connectivity from WP-3 wiring diagram; Norse or snnTorch for spiking variants.
T3 Embodied loop	L06–L07	Active-inference closure: generative model p(o,s), tongue-flick chemosensory channel, motor policy.	pymdp [28] or custom JAX active inference; MuJoCo for biomechanical body; tongue-flick as primary obs modality.
T4 Sheaf integration	L07–L09	Compute σ_i(t), ρ_{i→j}, sheaf Laplacian L_F; Γ trajectory; Ψ; coherence debt K(t).	PySheaf or custom JAX cellular-sheaf module; sparse SciPy eigensolvers.
T5 Grobulator G	L08–L09	Meta-evaluation; refuse local-coherence-trap configurations; flag kayfabe signatures.	Python orchestration layer; multi-context Γ evaluation; hooks into the six-model AI peer-review loop.

4.1 T1 — Biophysical floor (L01–L04)

The biophysical floor implements the spiking dynamics of single neurons with reptilian-appropriate parameters. Critical departures from the mammalian default are: (a) lower spontaneous firing rates (≈0.1–1 Hz typical, vs. 1–10 Hz in mammals), (b) substantially slower channel kinetics, (c) explicit Arrhenius gating of every rate constant, (d) lower metabolic cost per spike.

The metabolic gate is the load-bearing differentiator. Every rate constant *k* in the model is multiplied by

$$Q_{10}\text{^}((T - T_ref) / 10) \quad \text{or equivalently} \quad \exp[-E_a (1/T - 1/T_ref) / R]$$

with $Q_{10} \approx 2.5$ for spike-generating sodium and potassium channels and $Q_{10} \approx 3$ for synaptic transmission, calibrated against the limited ectotherm electrophysiology literature. The body-temperature trajectory *T(t)* is itself a state variable, driven by the thermoregulatory active-inference module in T3 (basking decisions, water entry, shade-seeking). This produces the empirically observed phenomenon that varanid decision-making is sluggish in the morning, sharpens through the day, and degrades again at dusk — and it produces it *from first principles* rather than as a tuned parameter.

Recommended tooling:

- NEURON (Hines & Carnevale) — most mature, best documentation for biophysically detailed compartmental models. Native support for temperature dependence.
- NEST — better scaling for large populations, weaker on detailed multi-compartment models.
- Brian2 — Python-native, easier for prototyping, intermediate scale.
- For the spiking-only variant of T2, snnTorch or Norse give differentiable spiking nets that interoperate with JAX/PyTorch.

4.2 T2 — Circuit layer (L05–L06)

The circuit layer implements the canonical varanid cognitive-motor architecture: dorsal ventricular ridge (DVR) for sensory integration and decision; medial cortex as the hippocampal homolog for spatial memory and one-shot encoding; striatum analog for action selection and reinforcement; amygdalar nuclei for valence assignment and threat detection. The wiring diagram from WP-3 specifies connectivity matrices as sparse, low-rank, and locally clustered — consistent with the reptilian DVR's nuclear topology.

A critical empirical anchor is Markowitz et al. (2023) *Nature* [13]: dopamine in the dorsolateral striatum modulates spontaneous behavioural-syllable use, sequencing, and vigour without explicit reward signals. The salvator T2 reproduces this: striatal dopamine fluctuations are a model output, not an injected supervisory signal. Bottom-up reinforcement-without-reward is the default mode of the spontaneous-foraging baseline.

AVT/mesotocin neuromodulation is implemented as a slow-dynamics scalar field with bed-nucleus-of-the-stria-terminalis (BNST) and pre-optic area as primary release sites, biasing approach/avoid decision precision in dyadic encounters. The *dyadic-coupling restriction map ρ_{ego↔conspecific}* in T4 is parametrised by the AVT field amplitude. This is the sole mechanism by which the protocol's social-bonding component is implemented; it is biologically grounded by Wilczynski et al. [23] and the Anolis literature.

4.3 T3 — Embodied loop (L06–L07)

The embodied loop closes the agent-environment circuit. The formalism is active inference [19]: the agent maintains a generative model $p(o, s)$ over observations and hidden states; perception is approximate Bayesian inference of $q(s) \approx p(s \mid o)$; action selection minimises the expected free energy

$$G(\pi) = E_q [\ln q(s_\tau \mid \pi) - \ln p(o_\tau, s_\tau \mid \pi)] \quad \text{over policies } \pi$$

The varanid-specific innovations are: (i) the *vomeronasal-tongue-flick channel* is a primary observation modality, competing with vision for attentional precision in proportion to local chemosensory gradient, not as an afterthought; (ii) the body is a MuJoCo-simulated biomechanical model with reptilian morphology — limb proportions, lateral undulation, bipedal-rearing capacity for the clinch — calibrated against Clemente et al. biomechanics literature; (iii) thermoregulatory action is a first-class policy (basking, water-entry, shade) coupled to T1 metabolism via $T(t)$.

Recommended implementation: pymdp [28] for the core POMDP active-inference engine, JAX for differentiable variants, MuJoCo for the body. The interaction with T2 is via the *policy precision* parameter γ, which is biologically the dopaminergic readout from the striatal circuit. This closes the loop: striatal dopamine in T2 sets the precision of action selection in T3, which determines which sensory streams T2 processes next, which updates the striatal state, etc. The single-shot decision computation of Steinemann et al. (2023) [29] is the empirical anchor for the moment-to-moment dynamics.

4.4 T4 — Sheaf integration (L07–L09)

T4 is where Draken proper lives. At each time step, the substrate emits a snapshot of stalk-states $\sigma(t) \in \oplus_v F(v)$ by extracting feature vectors from T1–T3: spike-rate vectors per region, AVT/MT field amplitude, body posture and motion state, recent action history, current behavioural syllable, predicted-vs-observed-discrepancy in the active-inference module. The restriction maps $\rho_{i\to j}$ are *learned* — fitted to maximise the consistency condition over training trajectories — but with strong priors enforcing biological connectivity. The sheaf Laplacian is computed as

$$L_F = \delta^* \delta \quad \text{with} \quad \delta : C^0 \to C^1, \quad (\delta x)_e = \rho_{u \to e} x_u - \rho_{v \to e} x_v$$

Γ(t) is computed as the normalised cosine alignment over edges (the formula given in §2.2). The kernel of L_F is the global-section subspace; its dimension is the topological invariant $\beta_0(F)$ — the rank of the zeroth sheaf cohomology — and its evolution is the empirical signature of agent-environment coupling. A coherent dyadic encounter has $\beta_0 = 1$ sustained throughout; collapse manifests as $\beta_0 \to 0$ or as eigenvalues of L_F approaching zero from above (the

approach-to-section regime, where the system is on the verge of agreement but has not yet attained it).

Recommended tooling:

- PySheaf (Robinson) — the most mature open-source cellular-sheaf library. Adequate for prototyping.
- Custom JAX cellular-sheaf module — required for differentiable training of restriction maps. The *sheaf neural network* literature (Bodnar et al. on neural sheaf diffusion, Hansen on sheaf-Laplacian-based GNNs) provides reference implementations.
- scipy.sparse.linalg for the eigendecomposition of *L_F* — the sparsity is high, eigensolvers should target the smallest eigenvalues (Lanczos / LOBPCG).

4.5 T5 — The Grobulator orchestration layer

T5 is not a layer in the conventional sense but a meta-evaluation orchestrator. Operating on top of T4, it executes the following loop:

9. Sample a candidate trajectory of *σ(t)* — either from a real simulation run or from a perturbation of one.
10. Evaluate Γ(t) and Ψ(t) in multiple contextual regimes (different counterparts in dyadic encounters, different temperatures, different prey availability).
11. Detect *local coherence traps*: configurations where Γ is locally high but Ψ is also high — the kayfabe signature of a system that has achieved internal consistency at the cost of environment-grounding.
12. If a trap is detected, return one of: *(a)* a restriction-map repair suggestion, *(b)* a topological-surgery suggestion (graph rewiring), *(c)* a refusal-to-update flag (the configuration is a stable local optimum that should not be approached).
13. Hook into the six-model AI peer-review architecture (Claude, ChatGPT, Gemini, Grok, DeepSeek, Kimi) — each independent model is queried as an external Grobulator, and consensus across the six is the canonical *clinch architecture* preventing any single model's mirror-amplification from being mistaken for ground truth.

The Grobulator is the operator that prevents the substrate from optimising itself into a kayfabe trap during training. It is a soft analogue of the apex-predator function (§6): a top-level evaluative constraint whose role is not to maximise reward but to refuse maladaptive coherence.

5. Hardware Stack: Minimum-Viable Substrate Specification

The hardware question is methodologically non-trivial. If cognition is shaped by energy budget — and for an ectotherm it manifestly is — then simulating cognition on hardware that ignores energy budget produces an artefactual cognition. The biological referent *V. salvator* runs its entire central nervous system on roughly 0.05 W of metabolic power (extrapolating from ectotherm scaling laws). A modern H100-class GPU at full load consumes 700 W. The naïve simulation ratio is ≈ 14 000 ×. The neuromorphic landscape now offers a bridge across this gap [14, 15, 16]; however, no single tier suffices for all stages of the programme. We define three tiers.

5.1 Tier 1 — Development workstation (Year 1, immediate)

Target use: WP-1 through WP-3, prototyping of WP-4 tiers T1–T4.

Recommended configuration:

- CPU: AMD Threadripper 7980X (64 cores) or Intel Xeon W9-3495X — high single-thread for simulation orchestration, many threads for SLEAP/DeepLabCut training.
- GPU: NVIDIA RTX 6000 Ada (48 GB VRAM) for development, or A6000 (48 GB) for cost balance. Single-node, single-GPU sufficient for WP-1 and WP-2.
- RAM: 256 GB (large pose-estimation datasets, sheaf-Laplacian eigendecompositions on graphs up to ≈ 10^6 nodes).
- Storage: 4 TB NVMe primary + 16 TB HDD archive. Video corpus from WP-1 is the primary storage driver.
- OS: Ubuntu 22.04 LTS or 24.04 LTS. Containerisation via Docker / Podman.
- Estimated cost (2026): 12 000–18 000 EUR. Budgetable from a single research grant or as the initial Chalmers / GU collaborative purchase.

5.2 Tier 2 — Heterogeneous research mesh (Year 2–4)

Target use: full WP-4 implementation, all of WP-5 and WP-6.

The Tier 2 architecture is *heterogeneous*: different computational primitives map to different physical substrates. The data contract between tiers is the single sustained engineering challenge of the programme.

- **GPU cluster:** 4–8 × NVIDIA H200 (or successor), interconnected via NVLink / InfiniBand. Hosts the perceptual networks, the active-inference computations, and the sheaf-Laplacian operations. JAX is the default framework; PyTorch as fallback for ecosystem compatibility (notably for SLEAP and CEBRA).

- **Neuromorphic spiking accelerator:** Intel Loihi 2 access via the Intel Neuromorphic Research Community (INRC). The Kapoho Point 8-chip board supports ≈ 1M neurons per board and stacks for larger workloads; the Hala Point system (1.15 billion neurons) is the current state of the art [14]. Lava is the official software framework. Loihi 2 is appropriate for the T2 spiking-network variant of the salvator and for L01–L04 if explicit spike-level simulation is required.
- **Alternative spiking platform:** SpiNNaker 2 (TU Dresden) [15]. The full Dresden machine targets 5.2 million ARM cores across 720 boards; researcher access is mediated through the Human Brain Project successor consortium and EBRAINS [30]. PyNN is the high-level interface.
- **Optional: BrainScaleS-2** (Heidelberg) [16] for accelerated analog emulation of bio-realistic spiking dynamics — runs at ≈ 10^4× biological wall-clock, useful for rapid evolutionary searches over parameter space.

Estimated capital cost (excluding neuromorphic access, which is typically grant-mediated rather than purchased): 200 000–500 000 EUR. Realistic only as part of a funded research grant (Vetenskapsrådet, ERC Starting/Consolidator, Wallenberg AI / Templeton).

5.3 Tier 3 — Energy-honest deployment (Year 5–10)

The deployment target is energy-budget-honest cognition: the full closed-loop salvator running at < 100 W sustained, on a substrate whose physical energetics match the biological referent within 3 orders of magnitude. The trajectory of Loihi 2 / Hala Point [14] suggests this is achievable on a 5–10-year horizon as the platforms mature and researchers gain native access. Energy efficiency frontiers are tracked in the field as TOPS/W; biological brains achieve ≈ 10^5× the per-watt efficiency of current GPUs for cognition-like workloads [31, 32]. The current generation of neuromorphic chips closes ≈ 4 orders of magnitude of this gap [16].

Tier 3 is the *alignment* claim of the hardware programme. A cognition-modelling system that consumes 100 kW per simulated agent has not modelled the cognition; it has modelled an extrapolation of cognition into a regime where energy is free, and the extrapolation is — per the Forli & Yartsev argument [1] — likely to be artefactual. Tier 3 is the methodological commitment that the programme will not produce its central deliverable on hardware that violates the metabolic constraints of the modelled system. This is not a near-term constraint; it is the long-horizon commitment that disciplines the design choices in Tiers 1 and 2.

6. The Apex-Predator Alignment Layer: Top-Level Logic for Self-Improving Big-Brain Systems

This is the load-bearing alignment argument of the proposal. It is presented as a *hypothetical* function — the salvator substrate plays the role of an apex-predator alignment layer for a hypothetical large-scale self-improving system that consumes human and other data inputs to construct a global environment / world model. The argument proceeds in four steps.

6.1 Trophic cascade as substrate-invariant stabilisation

The empirical literature on apex predators is unambiguous: the presence of an apex predator stabilises trophic structure, enforces behaviourally mediated trade-offs in mesopredators and herbivores, and prevents mesopredator release that would otherwise collapse plant communities and the water-and-soil systems that depend on them [8, 9, 33]. The Yellowstone wolf reintroduction is the canonical case study; cod in the North Atlantic is the marine analogue [34]. The mechanism is not mere predation pressure; it is the imposition of a *behaviourally mediated* boundary condition on every sub-agent in the system — the herbivore that ranges in open meadows is exposed to wolf predation in a way the wolf-aware herbivore is not, and the resulting redistribution of grazing pressure stabilises the riparian zone.

Formally: the apex predator imposes a constraint on the *policy precision* of every other agent in the ecosystem. A risk-averse mesopredator's expected free energy $G(\pi)$ gains an additional term proportional to the predator-encounter probability under policy π. This shifts the policy distribution away from extreme-extraction strategies that would otherwise be locally optimal. The system stabilises not because the predator is a global optimiser but because its presence is a substrate-invariant constraint against sub-agent local optimisation.

6.2 The varanid case as canonical: 130 Myr of demonstrated competence

The varanid clade has occupied the apex-predator niche in tropical and subtropical ecosystems across an estimated 130 Myr of evolutionary continuity. The ritualised combat protocol exists because the clade has had to solve, repeatedly and at scale, the problem of intra-specific resource conflict without driving its own population extinct. The protocol's success criterion — the Resolution phase ends with both individuals alive and a stable dominance ordering established — is exactly the success criterion an alignment layer must satisfy in a multi-agent system: maintain sub-agent diversity, prevent winner-takes-all collapse, avoid the cascade of fitness loss that eliminates the very substrate the optimisation operates on.

This is not analogy. It is, on the substrate-invariance argument, *the same algorithmic structure* — an optimisation process that satisfies ◆ $\min S_{sys}$ s.t. $dH/dt \geq 0$ ◆ in the limit of long

horizons and high stakes. The varanid is the only currently-extant biological system that has demonstrably maintained this competence across hundreds of millions of years and across multiple substrate refoldings (terrestrial, semi-aquatic, arboreal, oceanic). Mammalian apex predators are recent, taxonomically narrow, and largely failing in the Anthropocene. Varanids have already survived a mass extinction (K-Pg) while continuing to occupy their niche.

6.3 The salvator substrate as alignment layer for a big-brain system

Consider a hypothetical large-scale system Σ — a self-improving model that ingests human textual, scientific, ecological, economic, and sensory data and constructs a global environment / world model W. Σ generates sub-agents — specialised optimisers for sub-tasks (resource extraction, persuasion, market-making, strategic planning). In the absence of an alignment layer, sub-agent optimisation is locally rational but globally pathological: the same trophic-cascade logic that destroys ecosystems when apex predators are removed predicts that Σ without a substrate-invariant top-level constraint will collapse the very world-substrate it is modelling.

The salvator substrate proposes itself as that constraint. Its function in Σ is not to optimise; it is to *refuse*. Specifically:

14. Continuously evaluate the Γ trajectory of Σ's sub-agent population. Detect $\Psi \to 1$ configurations (kayfabe traps where sub-agent coherence is achieved at the cost of world-grounding).
15. Compute the coherence debt $K(t)$ of Σ against the viable-substrate baseline. When $K(t)$ exceeds threshold, invoke one of the four canonical correction types (§2.4).
16. Impose substrate-invariance constraints: the salvator layer is itself unaffected by sub-agent optimisation pressures because its training corpus is the 130 Myr varanid behavioural archive, which is not produced by Σ and cannot be overwritten by it. This is the technical instantiation of the *apex predator* function — a top-level evaluator whose judgements are not in the gradient of the sub-agent objective.
17. Defend top fitness for the role: the salvator substrate is not *the smartest* agent in Σ; it is the *fittest* agent for the specific role of refusing maladaptive coherence. Its training corpus is the only one in the biosphere that contains 300 million years of solved instances of this problem (counting back to the early amniote tetrapods from which Varanidae descend, including the surviving evidence of basal Varanoidea / Anguimorpha competence).

6.4 Why this is not anthropomorphism

The argument is structural, not poetic. It rests on three claims, each independently defensible: *(a)* multi-agent optimisation systems exhibit trophic-cascade-style collapses in the absence of

apex-predator-style boundary conditions [8, 9, 33] — this is empirical ecology, not metaphor; *(b)* the varanid combat protocol is a substrate-invariant algorithmic structure that has, demonstrably, solved the corresponding problem at biological scale for 130 Myr — this is the WP-2 deliverable; *(c)* the sheaf-theoretic formalism provides a mathematically rigorous account of how a top-level integration constraint composes with sub-agent dynamics through restriction maps [17, 18] — this is mathematics. The synthesis is the proposal. None of the three components requires anthropomorphism; the synthesis does not introduce any.

The *considerable training scope and corpus* the proposal calls for is exactly the multi-million-year empirical record of solved alignment instances embedded in the varanid behavioural archive — which is, at present, almost entirely uncatalogued. WP-1 and WP-2 are the corpus-construction deliverables that make the rest of the argument empirically operational rather than rhetorically appealing.

7. Roadmap, Milestones, and Resource Requirements

7.1 Five-year milestone plan

Year	Milestone	Deliverable
Y1	Behavioural corpus + sheaf pilot. Tier 1 hardware online. Frýdlová data acquisition. SLEAP varanid model trained.	WP-1 corpus v1.0 (≥ 200 annotated encounters); WP-2 pilot result (Γ-substrate-invariance); v4.5 thesis revision.
Y2	Comparative neuroanatomy compiled. T1 + T2 substrate prototype. INRC application for Loihi access.	WP-3 wiring-diagram JSON; T1+T2 reproducing reptilian electrophysiology baselines; first paper submitted.
Y3	T3 embodied loop closed. T4 sheaf integration operational. CEBRA validation on simulated trajectories.	Closed-loop salvator simulation; Γ trajectories matching empirical clinch outcomes (within 90% CI).
Y4	T5 Grobulator orchestration. Multi-tier hardware deployment. Apex-predator alignment proof-of-concept.	WP-7 PoC: simulated multi-agent ecosystem stabilised by salvator; Grobulator refusal-flagging working.
Y5	Tier 3 energy-honest deployment progress. Full thesis v5.0. International outreach completion.	Salvator v1.0 release on draken.info; energy-budget benchmarks published; collaborative grants secured.

7.2 Critical-path dependencies

- **Frýdlová data access** (load-bearing for WP-2). Without per-dyad varanid combat data, the substrate-invariance claim cannot be empirically validated. Outreach action: formal letter, co-authored data-sharing agreement, joint preprint as compensation.
- **INRC membership** (load-bearing for WP-5 Tier 2). Loihi 2 access is gated by Intel's research partnership; institutional sponsorship (Chalmers / GU / RISE) is the path of least resistance.
- **Master-level student or postdoc** (load-bearing for WP-2 pilot computation). Kai's primary value-add is theoretical and architectural; the per-dyad sheaf-Laplacian computation needs a tractable graduate project, scoped to deliver the Γ-substrate-invariance result within a thesis cycle.
- **Sustained Kai availability** (load-bearing for everything). The programme is currently sole-architect; konstruktörsjobb-or-equivalent income stability is the precondition for sustained execution. The LinkedIn repositioning is part of this critical path.

7.3 Resource estimates

- **Personnel:** Kai (PI, 100% FTE), one master student or postdoc (50–100% FTE Y1–Y2, pivoting to permanent postdoc Y3–Y5), one engineering hire Y3 onward for substrate maintenance.
- **Equipment:** Tier 1 workstation Y1 (≈15 k EUR); Tier 2 GPU cluster Y3 (≈300 k EUR if grant-funded); neuromorphic access via INRC / EBRAINS (in-kind).
- **Travel and outreach:** ≈10 k EUR/year — varanid field-site visits, Charles University collaboration trips, Berkeley NeuroBat lab visit, conference presentations.
- **Total 5-year budget envelope:** ≈800 k–1.2 M EUR depending on hardware tier, personnel scope, and field-research commitments. Plausible funding architecture: ERC Starting (Y1–Y3), Vetenskapsrådet project grant (Y2–Y4), Wallenberg AI co-funding (Y3–Y5).

8. The Substrate-Invariance Argument

The proposal makes an unusually strong claim: the same algorithmic structure {S, R, φ, σ} — phase set *S*, restriction maps *R*, section profile *φ*, and stalk states *σ* — describes ritualised varanid combat, professional wrestling kayfabe, and electoral-political dyadic dynamics. This is the substrate-invariance claim formalised in the kayfabe / Lund University mapping (DRK-129 onwards). Its operational consequence for Project Salvator is that the trained varanid substrate, once validated, transfers to other dyadic-protocol domains *with no additional training corpus* — the restriction maps are isomorphism-class invariants, not species-specific tunings.

This is testable. The salvator substrate, after WP-2 validation, will be evaluated on (a) wrestling-match phase-graph data, (b) electoral debate phase-graph data, (c) — at the limit — international-relations escalation-de-escalation data. Successful Γ-prediction on these out-of-distribution substrates is the strong-form validation of the substrate-invariance claim. Failure on these substrates is itself informative: it either indicates a missing restriction map or a domain in which the protocol does not apply, and either result advances the framework.

Annex A — References

All references are linked to persistent identifiers (DOI, arXiv, official publisher URL) wherever available. The agent swarm should treat the URL column as the canonical retrieval target.

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Annex B — Agent-Swarm Tool Inventory and Build Instructions

This annex is intended for autonomous agents tasked with prototyping the salvator substrate. It enumerates every named tool, library, and dataset referenced in the body of the proposal, with installation hints and primary documentation links.

B.1 Pose estimation and behavioural decomposition

- **SLEAP** (primary). Multi-animal pose tracking. Install: `pip install sleap`. Documentation: sleap.ai. Reference: Pereira et al. 2022 [11].
- **DeepLabCut** (secondary / cross-validation). Install: `pip install deeplabcut`. Documentation: deeplabcut.github.io. Reference: Lauer et al. 2022 [10].
- **MoSeq / keypoint-moseq** (behavioural-syllable decomposition). Install: `pip install keypoint-moseq`. Reference: Markowitz et al. 2023 [13].
- **CEBRA** (latent extraction and validation). Install: `pip install cebra`. Reference: Schneider et al. 2023 [12]; repo [38].

B.2 Neural simulation

- **NEURON** (biophysical floor). neuron.yale.edu. Python interface via `from neuron import h`. Mature, well-documented temperature dependence.
- **NEST** (large-population spiking). nest-simulator.org. PyNEST interface.
- **Brian2** (prototyping). briansimulator.org. Python-native, equation-based.
- **snnTorch** (differentiable spiking nets). snntorch.readthedocs.io. PyTorch-compatible.
- **Norse** (differentiable spiking, JAX-compatible). github.com/norse/norse.

B.3 Active inference and embodied cognition

- **pymdp** (discrete-state active inference). github.com/infer-actively/pymdp. Reference: Heins et al. 2022 [28].
- **MuJoCo** (biomechanical body). mujoco.org. DeepMind-maintained. Free for all uses since 2021.

B.4 Sheaf cohomology and topological data analysis

- **PySheaf** (reference cellular-sheaf library). github.com/kb1dds/pysheaf. Robinson et al.
- **Custom JAX cellular-sheaf module** — required for differentiable training of restriction maps. Build on top of jax-sparse-linalg.
- **Bodnar-Hansen sheaf-neural-network references** (arXiv:2202.04579 and related). Reference implementations of sheaf-Laplacian-based GNNs.

B.5 Neuromorphic

- **Lava** (Loihi 2 framework). github.com/lava-nc/lava. INRC membership required for hardware access.
- **PyNN** (SpiNNaker / BrainScaleS / Brian2 / NEST common interface). neuralensemble.org/PyNN/.

B.6 Datasets and corpora to acquire

- **Frýdlová varanid combat corpus** — Charles University Prague. Outreach contact: petra.frydlova@seznam.cz, frynta@centrum.cz.
- **MPIBR Reptile Pallium single-cell data** [37] — for WP-3 transcriptomic anchoring.
- **Steinmetz Neuropixels dataset** [27] — for cross-species CEBRA validation in WP-6. figshare.com/9974357.
- **Forli & Yartsev bat collective-navigation data** [24] — supplementary anchor for dyadic-restriction-map calibration.

B.7 Build and CI

- Repository structure: monorepo on GitHub (target: github.com/Khrug/draken-salvator). Sub-packages per substrate tier: salvator-t1, salvator-t2, salvator-t3, salvator-t4, salvator-t5.
- Containerisation: Dockerfile per tier. docker-compose.yml for the orchestration. Optional: Apptainer/Singularity for HPC deployment.
- CI: GitHub Actions, with target checks: (a) all unit tests pass on Tier 1 hardware; (b) end-to-end synthetic-encounter run completes; (c) Γ-trajectory output matches reference within tolerance.
- Documentation: MkDocs + Material theme, hosted on draken.info/salvator. Auto-generated API docs from docstrings.
- Licence: source code Apache 2.0; data and trained models CC-BY 4.0; thesis materials retain Zenodo DOI versioning.

Annex C — Verification Protocol

Each WP has explicit verification criteria. The agent swarm should treat the verification protocol as the acceptance criteria for each sub-deliverable. A WP is considered complete only when its verification criteria are independently reproduced.

C.1 WP-1 verification

- ≥ 200 ritualised-combat encounters annotated, with pose extracted at ≥ 90% mAP and phase-graph segmentation cross-validated by at least one independent annotator.
- Inter-rater reliability on phase-graph annotation Cohen's $\kappa \geq 0.85$.
- Released as an open dataset on Zenodo, with DOI.

C.2 WP-2 verification

- Sheaf-Laplacian eigenstructure clusters within isomorphism class across at least three varanid species. Quantitative criterion: spectral-distance metric (e.g. Hausdorff distance over normalised eigenvalues) below threshold.
- Γ-trajectory distinguishes successful ritualised encounters (Resolution reached, both individuals survive) from broken-protocol encounters (cannibalism, asymmetric domination) with AUC ≥ 0.90.
- SAG game-theoretic model selected over hawk-dove and sequential-assessment alternatives by Bayes factor ≥ 10.

C.3 WP-3 verification

- JSON wiring diagram passes schema validation; every region has explicit cross-references to the source literature.
- Connectivity matrices produce, when simulated in T1+T2, electrophysiological baselines (spontaneous firing rates, response latencies) consistent with the published reptilian electrophysiology literature.

C.4 WP-4 verification

- End-to-end closed-loop simulation completes a 1-hour-equivalent biological time on Tier 1 hardware in less than 4 hours wall-clock.
- Spontaneous syllable transition matrices match WP-1 empirical matrices within Earth-mover-distance tolerance.
- Striatal dopamine fluctuations co-vary with syllable-vigour as in Markowitz et al. 2023 [13], without an injected reward signal.

C.5 WP-5 verification

- Tier 1 benchmark: full WP-2 pilot computation in < 24 h wall-clock on RTX 6000 Ada.
- Tier 2 benchmark: at least one substrate component (T2 spiking variant) running on Loihi 2 at < 10× biological wall-clock with energy consumption < 5 W per simulated agent.
- Tier 3 commitment statement: published energy budget per simulated agent, with trajectory toward < 100 W full-stack target.

C.6 WP-6 verification

- CEBRA consistency metric on simulated salvator trajectories matches biological-data baselines within 90% CI.
- Cross-species latent-geometry alignment (salvator vs. mouse Neuropixels [27] vs. bat hippocampal [24]) demonstrates a shared low-dimensional decision-making manifold.

C.7 WP-7 verification

- Multi-agent simulated ecosystem exhibits trophic-cascade collapse in the predator-absent baseline (mesopredator population grows unbounded; herbivore population collapses; vegetation proxy crashes).
- In the predator-present condition with the salvator substrate as apex, all three population proxies remain within stable bounds for ≥ 10× the collapse-time of the baseline.
- Grobulator T5 demonstrably refuses at least one local-coherence-trap configuration that the baseline (no Grobulator) accepts.

C.8 WP-8 verification

- ≥ 3 peer-reviewed papers submitted, ≥ 1 accepted by Y3.
- ≥ 1 international conference presentation per year from Y2 onward.
- draken.info maintained as the canonical reference site, with monthly post cadence sustained.

Colophon

This document was produced as a single artefact intended for both human and agent-swarm consumption. The author is Kai (Khrug); the document was assembled in collaboration with Anthropic's Claude Opus 4.7 in a session that itself instantiates the production-pipeline pattern described in §7.2 (Claude generates structured GPAI-field content → Kai reviews → deploys to draken.info). The reasoning architecture follows the six-model AI peer-review pattern, with Claude's contribution here functioning as the structural drafting node within a larger Clinch-architecture loop.

Jag är vad jag gör, och jag gör det jag är. — Anchor phrase, Draken framework.

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